Hackathon Project Submission Checklist (Items 1 – 6)

Team Strawhats (Team Code: 627JUH) • Theme 2: Climate, Resource Infrastructure & Global Supply Resilience

1. PROBLEM STATEMENT

• Target Issue & Context: Macro weather alerts (~10 km city-wide averages) mask lethal micro-canyon heat traps while afternoon heat spikes trigger MSEDCL grid blackouts and municipal water cuts. Without constrained spatial deficit math, NGOs dispatch water tankers blindly and workers miss exertional CNS red flags (confusion/dizziness) and anhidrosis.

• Target Audience: (1) 10M+ Outdoor Gig & Transit Workers & (2) NGO Volunteer Leads & Municipal Disaster Response Teams.

2. SOLUTION DESCRIPTION

• High-Level Overview: Splits Pune into 64 micro-zones (Z-00 to Z-77, 3.3x3.3 km) ranked by deterministic Relief Deficit (Rd = Vh / (1 + Taps + 2×Clinics), where Vh = (Child_0-14 + Elderly_65+) × H38). Includes a Constrained Discrete Concave DP Allocator (verified vs brute force, 0.0% gap), Gemini 2.5 Flash multilingual dispatch with RefChecker 0.0% hallucination verification, and Vision anti-spoofing.

• Architecture: FastAPI + React 18 + Leaflet + SQLite powered by live Open-Meteo, WorldPop v2, and OSM Overpass.

3. PROTOTYPE LINK (LIVE DEPLOYMENT)

• Live Hosted Web App: https://heatatlas.vercel.app/

• Live API Readiness & Telemetry Probe: https://heatatlas.vercel.app/api/health

• Interactive OpenAPI Swagger Docs: https://heatatlas.vercel.app/docs

4. DEMO VIDEO (2–3 MIN END-TO-END LIVE DEMO)

• Official YouTube Demo Link: https://youtu.be/C0-BQNND-b8

• Direct Hosted MP4 Mirror: https://heatatlas.vercel.app/demo.mp4

• Highlights live AI reasoning, Constrained DP allocation (399 people served), 2-step CNS + Anhidrosis triage, Vision anti-spoofing & 3 portals.

5. GITHUB REPOSITORY LINK

• Public GitHub Repo: https://github.com/vishwjeetlol/heatatlas

• Includes clean README.md, specs.md, ARCHITECTURE.md, .env.example, prompts/, GitHub Actions CI, and 16/16 passing pytest acceptance tests (tests/test_heatatlas.py).

6. PRESENTATION / PITCH DECK (6-SLIDE PDF)

• Live Pitch Deck PDF: https://heatatlas.vercel.app/HeatAtlas_Pitch_Deck.pdf

• GitHub Deck Link: https://github.com/vishwjeetlol/heatatlas/blob/main/HeatAtlas_Pitch_Deck.pdf

• Submission Checklist PDF: https://heatatlas.vercel.app/HeatAtlas_Submission_Checklist.pdf